

Hennessey, J.L. and Patterson, D.A., 2016. Computer architecture: a quantitative approach. Elsevier. Chandramani Tiwary "Learning Apache Mahout", Packt Publishing, 2015. Fuchen Sun, Kar-Ann Toh, Manuel Grana Romy, KezhiMao, "Extreme Learning Machines 2013: Algorithms and Applications", Springer, 2014.
E-resources https://online.stanford.edu/courses/soe-ycs0007-mining-massive-data-sets https://www.edx.org/course/mining-massive-datasets https://www.my-mooc.com/en/mooc/mmds/ http://infolab.stanford.edu/~ullman/mmds/book.pdf
Topics relevant to "SKILL DEVELOPMENT": Hierarchical Clustering in a Euclidean and Non-Euclidean Space for Skill Development through Experiential Learning techniques. This is attained through assessment component mentioned in course handout.

Course Code: CSE3009	Course Title: Optimization Techniques for Machine Learning Type of Course: Discipline Elective in Artificial Intelligence and Machine Learning Basket Theory	L- T-P- C	3 -0	0	3
Version No.	1.0				
Course Pre-requisites	CSE3008 Machine Learning Techniques				
Anti-requisites	NIL				
Course Description	This course introduces a range of machine learning models and optimization tools that are used to apply these models in practice. Course will introduce what lies behind the optimization tools often used as a black box as well as an understanding of the trade-offs of numerical accuracy and theoretical and empirical complexity. For the students with some optimization background this course will introduce a variety of applications arising in machine learning and statistics as well as novel optimization methods targeting these applications.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Optimization Techniques for Machine Learning and attain Employability through Problem Solving Methodologies.				
Course Outcomes	On successful completion of this course the students shall be able to: Describe fundamentals of Machine learning [Knowledge]. Explain Machine learning models [Comprehension]. Discuss Convex optimization models [Comprehension].				

	Apply Methods for convex optimization [Application].			
Course Content:				
Module 1:	Fundamentals of Machine learning	Quiz	Knowledge based Quiz	8 Sessions
Topics: Machine learning paradigm, empirical risk minimization, structural risk minimization, learning guarantees, introduction of VC-dimension.				
Module 2:	Machine learning models	Quiz	Comprehension based Quiz	10 Sessions
Topics: logistic regression, support vector machines, sparse regression, low dimensional embedding, low rank matrix factorization, sparse PCA, multiple kernel learning.				
Module 3	Convex optimization models	Assignment	Batch-wise Assignments	9 Sessions
Topics: linear optimization, convex quadratic optimization, second order cone optimization, semidefinite optimization, convex composite optimization				
Module 4:	Methods for convex optimization	Assignment and Presentation	Batch-wise Assignment and Presentations	11 Sessions
Topics: gradient descent, Newton method, interior point methods, active set, prox methods, accelerated gradient methods, coordinate descent, cutting planes, stochastic gradient.				
Targeted Application & Tools that can be used: Use of Matlab tool				
Project work/Assignment: Survey on Methods for convex optimization				
Text Book T1. Charu C. Aggarwal, “ Linear Algebra and Optimization for Machine Learning”, Springer, 2020. T2. Sra Suvrit, Nowozin Sebastian, and Wright Stephen J, “Optimization for Machine Learning”, The MIT Press, 2012.				
References R1. Guanghui Lan, “First-order and Stochastic Optimization Methods for Machine Learning”, Springer Cham, 2020. Web References W1. https://sm-nitk.vlabs.ac.in/ W2. https://nptel.ac.in/courses/				
Topics related to development of “EMPLOYABILITY SKILL”: Convex optimization models and Methods for convex optimization, for development of Employability Skills through Participative				

Learning Techniques. This is attained through assessment components mentioned in course handout.